

Prodigy Pal

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Chapter 1

Introduction

ProdigyPal is a website application that is targeted towards teachers, students and educational institutes. It aims to generate dynamic and interactive worksheets for students to solve and learn from. ProdigyPal will keep track of how well the students perform in different topics and provide a detailed assessment to their teachers. It will also provide the students with questions that are personalized and suited to their difficulty level, and targets topics they are both weak and strong in. The software will include advanced features such as automated student diagnostics for teachers and parents/guardians, personalized worksheet generation, leaderboards and achievements to motivate engagement, and an intelligent chatbot that would provide real-time assistance to both students and teachers.

This document outlines the requirements and specifications of the ProdigyPal application. It is intended for a wide range of readers who will be involved or impacted by the development and use of the application:

1. **Developers:** Require detailed specifications to assist them with ProdigyPal's features and technical requirements.
2. **Project Managers:** Will use the documentation to understand the scope, timelines and deliverables associated with the development and deployment of ProdigyPal.
3. **Marketing Staff:** To gain insights into core features and the Unique Selling Points of ProdigyPal.
4. **Users:**
 - (a) **Teachers:** Required to know how the tool works to simplify their workloads.
 - (b) **Students:** Compete with themselves and their peers on achieving more correct results in different topics.
 - (c) **Educational Institutions:** Focus on educational objectives and learning outcomes.

5. **Testers:** Utilize the document to design test cases and ensure that ProdigyPal meets both functional and non-functional requirements.
6. **Documentation Writers:** For creating user guides, FAQs, and onboarding materials that supports teachers and students in using the platform.

1.1 Problem Statement

In traditional educational settings, teachers often lack the flexibility to tailor learning materials to individual student needs. They are constrained by standardized resources that do not account for the diverse learning paces, skill levels, or familiarity students have with the content. As a result, students who struggle with certain topics risk falling behind, while those who excel may not be sufficiently challenged. These limitations hinder personalized learning and engagement, creating a one-size-fits-all approach that fails to address unique student needs. ProdigyPal is being developed to solve this problem by providing a platform that generates customized, dynamic worksheets. This software will also allow teachers to offer personalized learning experiences, track student performance, and adjust to individual strengths and weaknesses. By creating an adaptive learning environment ProdigyPal aims to uplift struggling students while challenging high achievers, in turn fostering a more personalized and effective learning process.

1.2 Scope

The scope of ProdigyPal includes the development of a web-based application designed to generate personalized, dynamic worksheets for students in grades 1 to 6, focusing on mathematics. The system will use machine learning models to create adaptive questions tailored to individual student performance. It will track student progress and offer teachers detailed performance analytics, helping them identify strengths and areas for improvement. The application will feature gamification elements like leaderboards and achievements to keep students engaged and motivated. Teachers and students will have different access roles, with features customized to their needs. ProdigyPal aims to provide a scalable and user-friendly platform that enhances the learning experience by offering tailored educational content and real-time assistance. The project covers the full range of worksheet generation, from data processing to final delivery, within a reliable and adaptive system.

1.3 Modules

1.3.1 Module 1: Dataset and Data Pipeline

This module focuses on building and maintaining reliable datasets that the Machine Learning models can use for generating content. It also handles the flow of data from the database through the different models and into the web application, ensuring seamless data processing.

1. **Dataset Collection and Preparation:** Collects and processes raw data needed for training the models.
2. **Data Pipeline Management:** Manages the flow of data from the database to the ML models and ensures its availability for real-time processing.

1.3.2 Module 2: Machine Learning Models

This module is dedicated to the design, development, and training of the machine learning models that power ProdigyPal's personalized learning experience.

1. **Question Generation Model:** Dynamically generates adaptive questions based on students' past performance and current difficulty levels.
2. **Image Generation Model:** Produces relevant visual aids, such as diagrams or illustrations, to accompany the generated questions.
3. **Answer Explanation Model:** Provides step-by-step solutions to the generated questions, helping students understand how to arrive at the correct answers.

1.3.3 Module 3: Worksheet Assembly and Integration

This module oversees the integration of the different ML-generated components, creating a cohesive and dynamic worksheet that can be accessed by users through the web application.

1. **Content Organization:** Organizes the questions, images, and answers generated by the models, labeling them as worksheet assets.
2. **Worksheet Compilation:** Combines all assets (questions, images, solutions) into a final, structured worksheet ready for student interaction.

1.3.4 Module 4: Web Application

This module deals with the development and user interface of the web application, which provides teachers, students, and other stakeholders access to ProdigyPal's features.

1. **Dynamic Worksheet Delivery:** Presents the personalized worksheets to students, allowing them to solve and submit responses in real time.
2. **Student Analytics Dashboard:** Allows teachers to track and analyze student performance, providing insights into learning progress across various topics and difficulty levels.

User class	Description
Student	Students are the primary users who will interact with the personalized worksheets generated by ProdigyPal. They will solve questions and receive immediate feedback, while their performance data is tracked and analyzed to help identify strengths and areas for improvement.
Teacher	Teachers are responsible for overseeing the generation of customized worksheets, selecting topics and difficulty levels based on individual student needs. They will monitor student progress and use detailed performance analytics to provide targeted guidance and support in line with educational goals.
Educational Institute	Educational Institutes establish the learning objectives and standards that Teachers must follow. They have administrative oversight of the content being generated, ensuring that worksheets align with institutional policies and curricula. Additionally, they can review teacher performance and the overall effectiveness of the learning materials.
Parent/ Guardian	Parents or Guardians will have access to their child's performance reports and progress analytics. They can monitor their child's learning journey, track their improvement over time, and engage with teachers to support personalized learning strategies.

1.4 User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

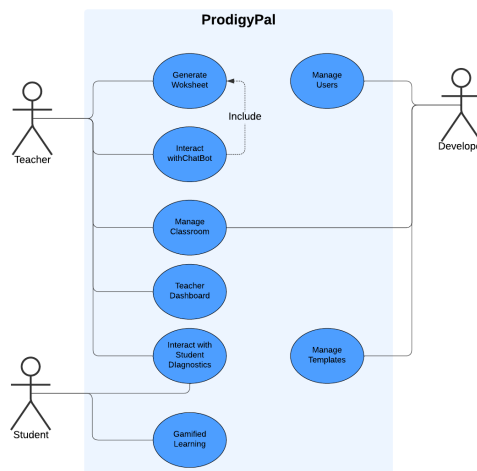


Figure 2.1: Use Case Diagram

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depend on

what kind of technique has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes. ?

2.2.1 Module 1: Worksheet Generation

Following are the requirements for module 1:

- FR-1:** The system shall allow teachers to select specific topics and difficulty levels for each student or group of students.
- FR-2:** The question generation ML model shall generate questions based on the specified topics and adjust for the student's performance levels.
- FR-3:** The system shall integrate relevant images alongside the generated questions to create fully-formed worksheets.
- PER:** The system shall generate a fully personalized worksheet in less than 1 minute for individual students and within 10 minutes for a class of up to 30 students.

2.2.2 Module 2: Automated Student Analytics

Following are the requirements for module 2:

- FR-1:** The system shall automatically record each student's performance after they complete a worksheet.
- FR-2:** The system shall generate diagnostic reports for teachers, highlighting students' strengths and weaknesses based on performance data.
- FR-3:** The system may provide teachers with recommendations for additional resources or targeted worksheets for struggling students.
- PER:** The system shall update student performance metrics and generate a diagnostic report within 10 seconds of completing a worksheet.

2.2.3 Module 3: Dynamic Image Generation

Following are the requirements for module 3:

- FR-1:** The system shall generate relevant images based on the content and variables of the questions.

FR-2: The image ML model shall dynamically adjust the visual content according to the parameters in the questions.

FR-3: The images should seamlessly integrate into the worksheet layout to enhance interactivity.

2.2.4 Module 4: Role-Specific Dashboards

Following are the requirements for module 4:

FR-1: The teacher dashboard shall present detailed analytics on student performance, recommended learning plans, and overall class progress.

FR-2: The student dashboard shall display personalized worksheets, performance feedback, and achievements to encouraging learning progress.

FR-3: The dashboard interface shall allow easy navigation for both students and teachers to move between different sections.

2.2.5 Module 5: Gamified Elements

Following are the requirements for module 5:

FR-1: The system shall include leaderboards displaying student rankings based on performance metrics.

FR-2: The system shall allow teachers to set achievement goals for students.

FR-3: The system shall notify students about their progress on leaderboards and achievements, encouraging engagement and motivation through badges or rewards.

2.2.6 Module 6: AI Chatbot Assistance

Following are the requirements for module 6:

FR-1: The chatbot shall assist students by providing step-by-step guidance for solving problems when requested

FR-2: The chatbot shall help teachers with queries related to platform features or provide insights into student performance and progress.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable. The following are some examples of documenting guidelines.

2.3.1 Reliability

- REL-1:** The system shall achieve an MTBF of at least 1,000 operational hours under normal usage conditions. A failure is defined as any incident where a user cannot generate worksheets, track student performance, or access essential features for over 30 minutes.
- REL-2:** In case of a failure, no more than 5% of user-generated data shall be lost or corrupted. The system shall maintain transactional integrity, ensuring that either all or none of the data associated with an operation is saved.
- REL-3:** In the event of a failure, the system shall recover and resume normal operations within 10 minutes, including restoring access to all core features.

2.3.2 Usability

- USE-1:** The system shall ensure that new users can learn to navigate the core features within 15 minutes of their first interaction, without external assistance.
- USE-2:** At least 90% of tasks shall be completed by users without needing more than 5 clicks or interactions per task.
- USE-3:** The system should provide contextual tooltips and inline validations to prevent users from making common errors
- USE-4:** The system shall provide clear and actionable error messages that include descriptions of what went wrong and how to fix it. Users should be able to recover from common errors (e.g. incorrect worksheet topic selection) within 1 minute.
- USE-5:** Teachers shall be able to generate a new personalized worksheet for a student or class in less than 2 minutes, with minimal input required beyond topic selection and difficulty adjustment.

2.3.3 Performance

- PER-1:** All pages shall load fully within 4 seconds under normal server load conditions.
- PER-2:** The system shall retrieve student performance records and worksheet history from the database within 10 seconds for queries involving up to 100 students.
- PER-3:** The system shall be capable of supporting 1,000 concurrent users with no noticeable degradation in response times (no more than a 10% increase in worksheet generation time or chatbot response time).

2.3.4 Security

- SEC-1:** The system shall automatically log out users after 15 minutes of inactivity to prevent unauthorized access through an unattended session.
- SEC-2:** The system shall enforce role-based access control (RBAC), ensuring that users can only access resources and data necessary for their roles.
- SEC-3:** The system shall maintain detailed audit logs of all user activities, including login attempts, data access, and changes made to user profiles. These logs should be immutable and retain data for at least 12 months.
- SEC-4:** The system shall implement an authentication mechanism for all user accounts to prevent unauthorized access.

2.4 Domain Model

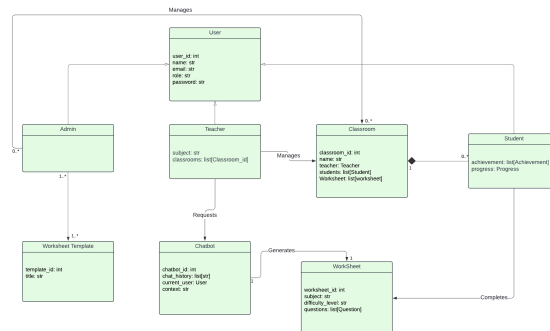


Figure 2.2: Domain Model

Chapter 3

System Overview

ProdigyPal is a personalized learning solution enhanced by AI, catering to both teachers and students to enrich the educational journey. The objective is to tackle the deficiencies of conventional learning methods, which frequently lack flexibility and adaptability, resulting in inconsistent student results. The main emphasis of the platform is on Mathematics for students up to Grade 6, providing a flexible and interactive space for learning and evaluation

ProdigyPal offers two main features for two primary users: **For Teachers:** It provides a smart chatbot with AI capabilities to help create interactive worksheets. These worksheets are customized to cater to the unique requirements and pace of learning of each student, enabling teachers to offer personalized learning resources. It consists of resources for monitoring student achievements, providing beneficial information on each student's advancement, and aiding teachers in filling learning deficiencies.

For Students: It provides an interactive learning experience that enhances participation and drive. By including leaderboards, point systems, and prompt feedback, students are motivated to engage in their learning process. The system offers adaptive learning pathways, giving students challenges tailored to their skill levels, as well as providing hints and feedback to support their progress.

ProdigyPal utilizes advanced technologies like machine learning models, dynamic content generation, and real-time data analytics to create a personalized and interactive learning environment for students, giving teachers more flexibility and control over educational resources.

3.1 Architectural Design

ProdigyPal’s architectural design adopts a modular strategy, dividing the system’s functionality into separate but interconnected components. This design improves scalability and maintainability, while also enabling separate development, testing, and deployment of each module.

The system is divided into four main modules, with each one handling a particular aspect of the platform’s functioning:

3.1.1 Dataset and Data Pipeline Module

This module is in charge of collecting, standardizing, and organizing the data needed to train machine learning models. It establishes a data pipeline to guarantee smooth data transition from capturing to analyzing.

The module serves as a base, as the information gathered here is used in the ML training module to guarantee that the worksheets created are precise and customized for the students’ requirements.

3.1.2 Machine Learning (ML) Models Training Module

This module is responsible for the training and adjusting of machine learning models. It is in charge of creating dynamic worksheets that range in difficulty levels according to students’ skills.

Essential components like engineering and the Chatbot backend take inputs from teachers and convert them into customized, AI-generated materials.

The result of this module seamlessly connects with the front-end to create worksheets in real-time.

3.1.3 Client Web Application Module

This module serves as the interface that teachers and students engage with. Constructed using ReactJS, which enables an interactive and adaptable user interface.

Teachers have the option to utilize the Custom Worksheet Generation Prompter for creating materials, as students engage with the gamified UI that features progress tracking and engagement resources.

This component interacts with the back-end APIs, obtaining information produced by the

ML models and displaying it in an easy-to-use layout.

3.1.4 Testing Module

It hosts the datasets, ML models and user-facing features in various stages of development to ensure full testing is being carried out on all during the entire process. It validates that making worksheets and grading kids with their game-like counterparts is really awesome. Since the rest of the modules test concurrently to ensure a seamless test experience as well as ensuring no unexpected failures are observed.

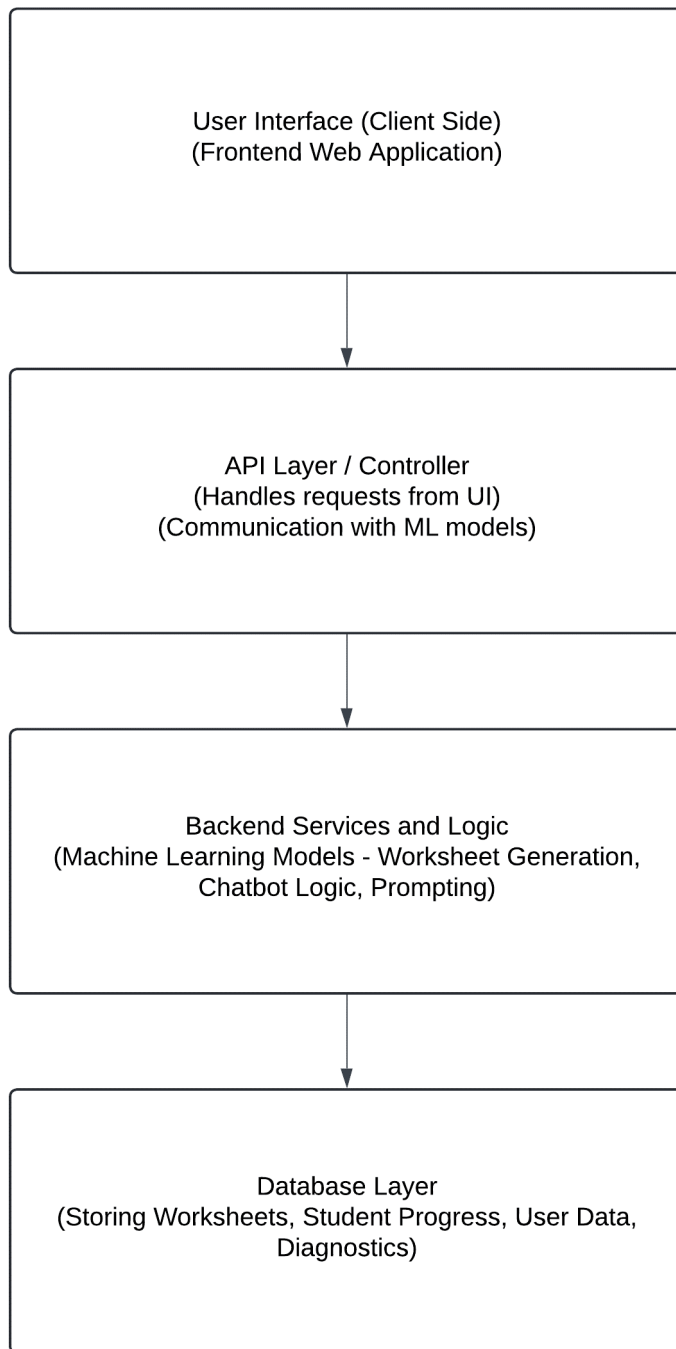
3.1.5 Relationships Between Modules

The data pipeline is essential for the system, providing necessary data to the ML training module, which produces content using the data. The Client Web Application serves as the platform for teachers and students to engage with the content produced. Real-time feedback and analytics are sent back to the system, enabling continuous enhancements to the worksheets and diagnostics. The Testing module verifies that all interactions among these modules operate as planned.

This modular design ensures different areas of focus are kept separate, making the system simple to handle, upgrade, and expand. Every module has a unique task but stays linked to attain dynamic learning, personalized education, and teacher empowerment functionalities.

3.1.6

Initial Design Stage The high-level structure of the ProdigyPal is shown through a basic Box and Line Diagram. This diagram highlights the primary subsystems and their relationships, while avoiding detailed discussion of the specific architectural style at this point.



3.2 Design Models

Create design models as are applicable to your system. Provide detailed descriptions with each of the models that you add. Also ensure visibility of all diagrams.

Design Models for Object Oriented Development Approach

3.2.1 Activity Diagram

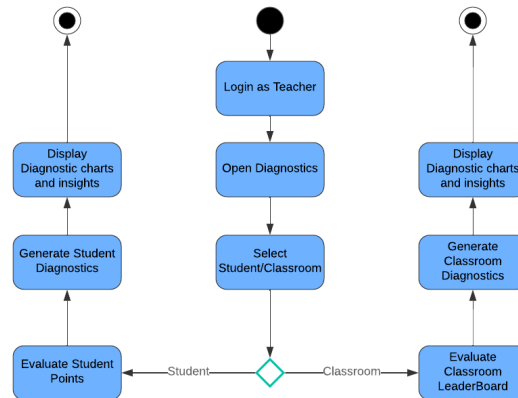


Figure 3.1: Activity diagram - Classroom Students Diagnostics

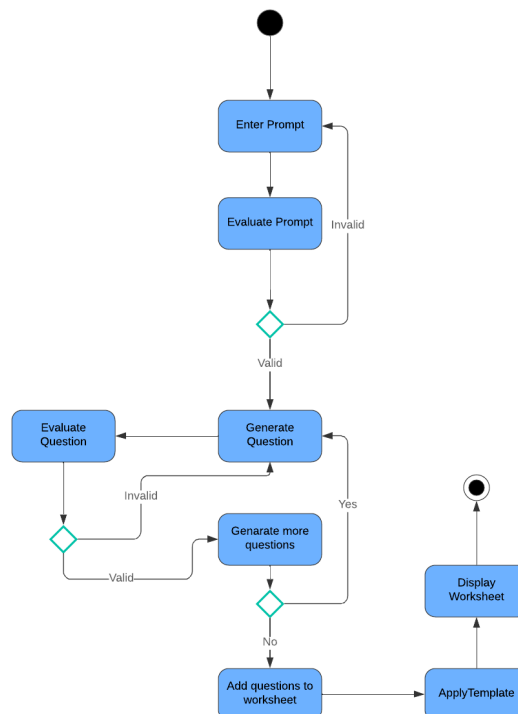


Figure 3.2: Activity diagram - Generate Worksheet

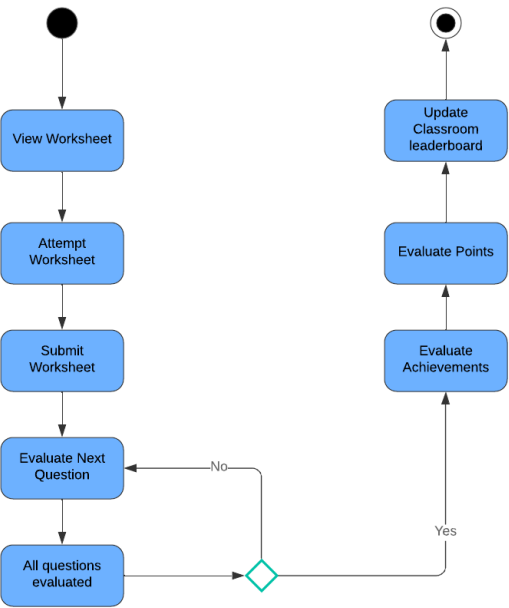


Figure 3.3: Activity diagram - Student Worksheet Attempt

3.2.2 Class Diagram

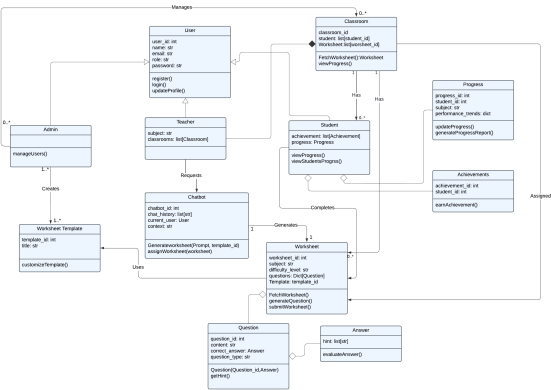


Figure 3.4: UML Diagram

3.2.3 Class-Level Sequence Diagram

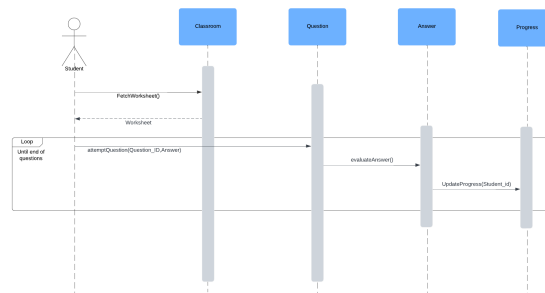


Figure 3.5: Sequence diagram - Student Worksheet Attempt

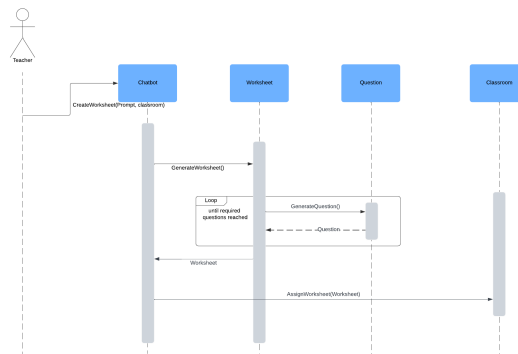


Figure 3.6: Sequence diagram - Worksheet Generation

